

CHI-WEI HU

📍 California, CA | 📞 (626)494-6280 | ✉️ chiwei@usc.edu | 🔗 <https://www.linkedin.com/in/william-hu-922261247/>

Aspiring software engineer currently pursuing a MS in Computer Science at USC. Proficient in C#, C++, and Python, with experience in establishing automation process, database, web and iOS App development. Actively seeking a software engineer internship position.

Education

University of Southern California(USC)

Los Angeles, U.S.

MS in Computer Science

Sept. 2024 - Present

- Coursework: Database Systems (CSCI585); Programming Systems Design in Java (CSCI455)

Taiwan University (NTU)

Taipei, Taiwan

MS in Biochemistry

Sept. 2020 - July2022

- Thesis: Engineering of A Multi-cell Type Immunotherapy leveraging synNotch-driven Macrophages
- GPA: 3.88/4.3

National Cheng Kung University (NCKU)

Tainan, Taiwan

BA in Life Science

Sept. 2017-July 2020

- Research project: Organelle stress effect of Estrogen metabolites 4-Hydroxyestradiol and 2-Hydroxyestradiol in Astrocytes

Skills

- **Languages:** C#, C++, Python, Java/Open JDK, XAML, HTML, R
- **Tools:** MySQL, MongoDB, Command line interface (Bash/ Linux OS), Git, Firmware, RS-232/RS-485 Communication protocol, SEMI Equipment Communication Standard (SECS) Communication protocol, CSS, SwiftUI, System integration, Automation process, Embedded system
- **Framework:** WPF, Express.js, Flask
- **Coursework:** Web Development bootcamp (python/HTML); Data Structure and Algorithm (C++) (A-); Artificial Intelligence (A+); Introduction to Bioinformatics (R) (A); Introduction and Practice of Databases (PHP, MySQL) (A)

Experience

Software Engineer, Department of Computer-Integrated Manufacturing (CIM)

Tainan, Taiwan

Strong Electro-Mechanical Co., Ltd

Sept. 2022- May 2024

- Developed a Standalone Automation Process for etching machines in panel factory in C#, utilizing SECS for device communication and improving process efficiency by 20%
- Established a C++ monitoring system for exposure machines in a panel factory, enabling communication with embedded systems like Arduino and DAQ devices, and integrating with MySQL, resulting in a 15% reduction in downtime
- Served as main liaison with clients during the testing phase, providing real-time support to collect feedback and coordinate with team members to iteratively optimize the codes before official launched

Intern, Department of Computer-Integrated Manufacturing (CIM)

Tainan, Taiwan

Strong Electro-Mechanical Co., Ltd

July 2018-Aug. 2018

- Rewrote a C++ vehicle weighing system in C#, with frontend developed using WPF framework and backend integrated with MySQL database
- Leveraged the SMTP communication protocol to implement automatic email notification, enhancing system performance and user communication

Projects & Research Experience

Movie Check Mobile App [\(Link\)](#)

- Built a customized movie tracking app designed with SwiftUI, handling the API to fetch the latest information and media content of TV shows on TMDB website
- Leveraged Express.js to connect to MySQL database, enabling rolling tracking and chronological statistics of viewing habits

Personal Portfolio Website [\(Link\)](#) [\(Github\)](#)

- Built a personal portfolio website designed with HTML and CSS
- Deployed flask and jinja2 in python as backend development and MongoDB as database to store information

Master Student, Institute of Biochemistry | Advisor: Steven Lin

Taipei, Taiwan

Academia Sinica

Apr. 2020-June 2022

- Established recombinant protein bank by conducting protein structure analysis and sequence alignment through BLAST and extensive literature review
- Utilized Lenti-transfection to introduce the synNotch tool into various immune cell types, thoroughly assessing its functionality via flow cytometry for over 50% expression efficiency
- Conducted Ni-NTA protein purification for cytotoxic antibodies and verified for 20% cytotoxic ability against solid tumors by ADCC cytotoxicity assay